

HYBRID TOPOLOGY

1. Project Overview:

This project implements a hybrid topology while taking into account the principles of both the ring topology and star topology connected together by router. The purpose of the design is to demonstrate VLAN segmentation, inter-routing, IPv4 and IPv6 configs with basic security on Cisco Packet Tracer.

2. Network Design Summary:

- Ring topology: It consists of 5 departments (vlan 10-50) each having its designated switch that further branches out to the Multilayered switch in the center of the design. The access switches all have a pair of two computers connected to them.
- Star: Various servers can be found there, Email, HTTP, DNS and DHCP, in which the DHCP server provided the entire system with ipv6 addresses
- Router: Responsible for routing between the server farm and the ring
- ACL: Provided basic security by inhibiting unwanted VLANs.

3. VLAN Configs

VLAN ID	Department	IPv4 network	IPv6 network
10	Admin	192.168.10.0/24	2001:DB8:10::/64
20	HR	192.168.20.0/24	2001:DB8:20::/64
30	IT	192.168.30.0/24	2001:DB8:30::/64
40	Finance	192.168.40.0/24	2001:DB8:40::/64
50	Legal	192.168.50.0/24	2001:DB8:50::/64
60	Sever Farm	192.168.60.0/24	2001:DB8:60::/64

4. Device Summary:

Device	Function	Key Configuration
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Multilayered switch (MLS)	Host VLAN 10-50 and does inter VLAN-Routing of ipv4 only	Vlan creation, trunk t router, IP routing enabled.
Router	Connect both topologies together and provide pathway.	Sub-interfaces for VLANS, ACL for control and security as well as IPv4/IPv6
Server	DNS, HTTP, Email and DHCP are hosted there	Dual-stack IPv4/IPv6
PCs	End devices with 10-50 VLAN	IPv6 and v4 addressing
Access switches	Forwards packets sent	Access and trunk ports

5. Configuration:

Configs on the MLS

Switch# configure terminal

Switch(config)# vlan 10

Switch(config-vlan)# name HR

Switch(config-vlan)# exit

Switch(config)# vlan 20

Switch(config-vlan)# name IT

Switch(config-vlan)# exit

Switch(config)# vlan 30

Switch(config-vlan)# name Finance

Switch(config-vlan)# exit

Switch(config)# vlan 40

Switch(config-vlan)# name Admin

Switch(config-vlan)# exit

```
Switch(config)# vlan 50
```

```
Switch(config-vlan)# name Sales
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 60
```

```
Switch(config-vlan)# name ServerFarm
```

```
Switch(config-vlan)# exit
```

6. Trunk config between mls and router :

```
Switch(config)# interface gig0/1
```

```
Switch(config-if)# switchport trunk encapsulation dot1q
```

```
Switch(config-if)# switchport mode trunk
```

```
Switch(config-if)# switchport trunk allowed vlan 10,20,30,40,50,60
```

```
Switch(config-if)# exit
```

7. Router sub-interfaces for inter-vlan routing

```
Router(config)# interface gig0/0/0.10
```

```
Router(config-subif)# encapsulation dot1Q 10
```

```
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
```

```
Router(config-subif)# ipv6 address 2001:DB8:10::1/64
```

```
Router(config-subif)# no shutdown
```

```
Router(config-subif)# exit
```

```
Router(config)# interface gig0/0/0.20
```

```
Router(config-subif)# encapsulation dot1Q 20
```

```
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

```
Router(config-subif)# ipv6 address 2001:DB8:20::1/64
```

```
Router(config-subif)# no shutdown
```

```
Router(config-subif)# exit
```

```
Router(config)# interface gig0/0/0.30

Router(config-subif)# encapsulation dot1Q 30

Router(config-subif)# ip address 192.168.30.1 255.255.255.0

Router(config-subif)# ipv6 address 2001:DB8:30::1/64

Router(config-subif)# no shutdown

Router(config-subif)# exit
```

```
Router(config)# interface gig0/0/0.60

Router(config-subif)# encapsulation dot1Q 60

Router(config-subif)# ip address 192.168.60.1 255.255.255.0

Router(config-subif)# ipv6 address 2001:DB8:60::1/64

Router(config-subif)# no shutdown

Router(config-subif)# exit
```

Enabling routing on the router

```
Router(config)# ip routing

Router(config)# ipv6 unicast-routing
```

8. ACCES CONTROL LIST CONFIG (basic security) :

```
Router(config)# access-list 100 permit tcp any any eq www

Router(config)# access-list 100 permit tcp any any eq domain

Router(config)# access-list 100 permit ip 192.168.10.0 0.0.0.255 192.168.60.0 0.0.0.255

Router(config)# access-list 100 permit ip 192.168.20.0 0.0.0.255 192.168.60.0 0.0.0.255

Router(config)# access-list 100 permit ip 192.168.30.0 0.0.0.255 192.168.60.0 0.0.0.255

Router(config)# access-list 100 permit ip 192.168.40.0 0.0.0.255 192.168.60.0 0.0.0.255

Router(config)# access-list 100 permit ip 192.168.50.0 0.0.0.255 192.168.60.0 0.0.0.255

Router(config)# access-list 100 deny ip any 192.168.60.0 0.0.0.255

Router(config)# exit
```

9. ACL leading to the farm

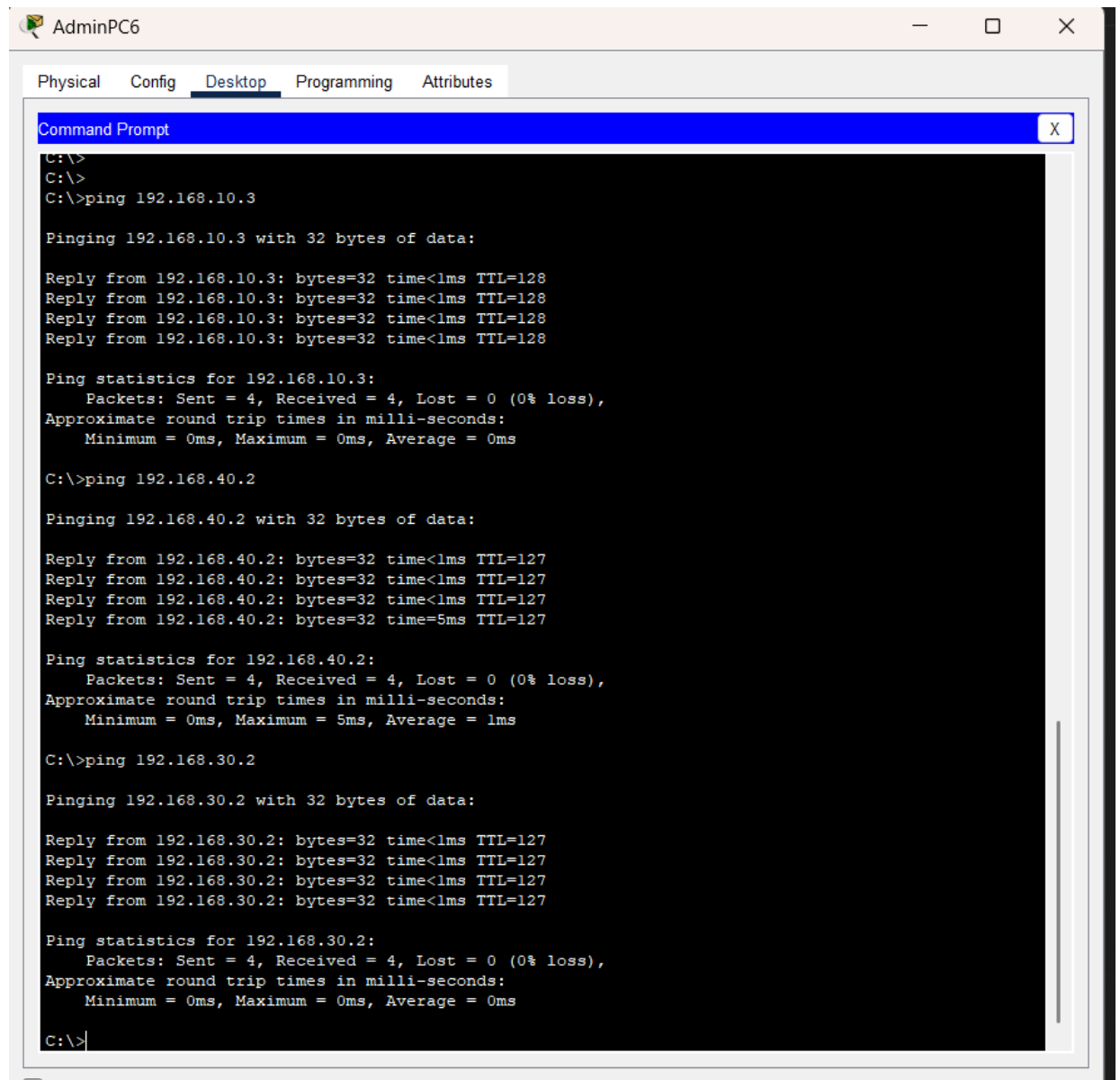
```
Router(config)# interface gig0/0/0.60

Router(config-subif)# ip access-group 100 in
```

Router(config-subif)# exit

10. Testing and verification:

a) Ping between and outside of VLANs:



```
AdminPC6
Physical Config Desktop Programming Attributes
Command Prompt
C:\>
C:\>
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.40.2

Pinging 192.168.40.2 with 32 bytes of data:

Reply from 192.168.40.2: bytes=32 time<1ms TTL=127
Reply from 192.168.40.2: bytes=32 time<1ms TTL=127
Reply from 192.168.40.2: bytes=32 time<1ms TTL=127
Reply from 192.168.40.2: bytes=32 time=5ms TTL=127

Ping statistics for 192.168.40.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

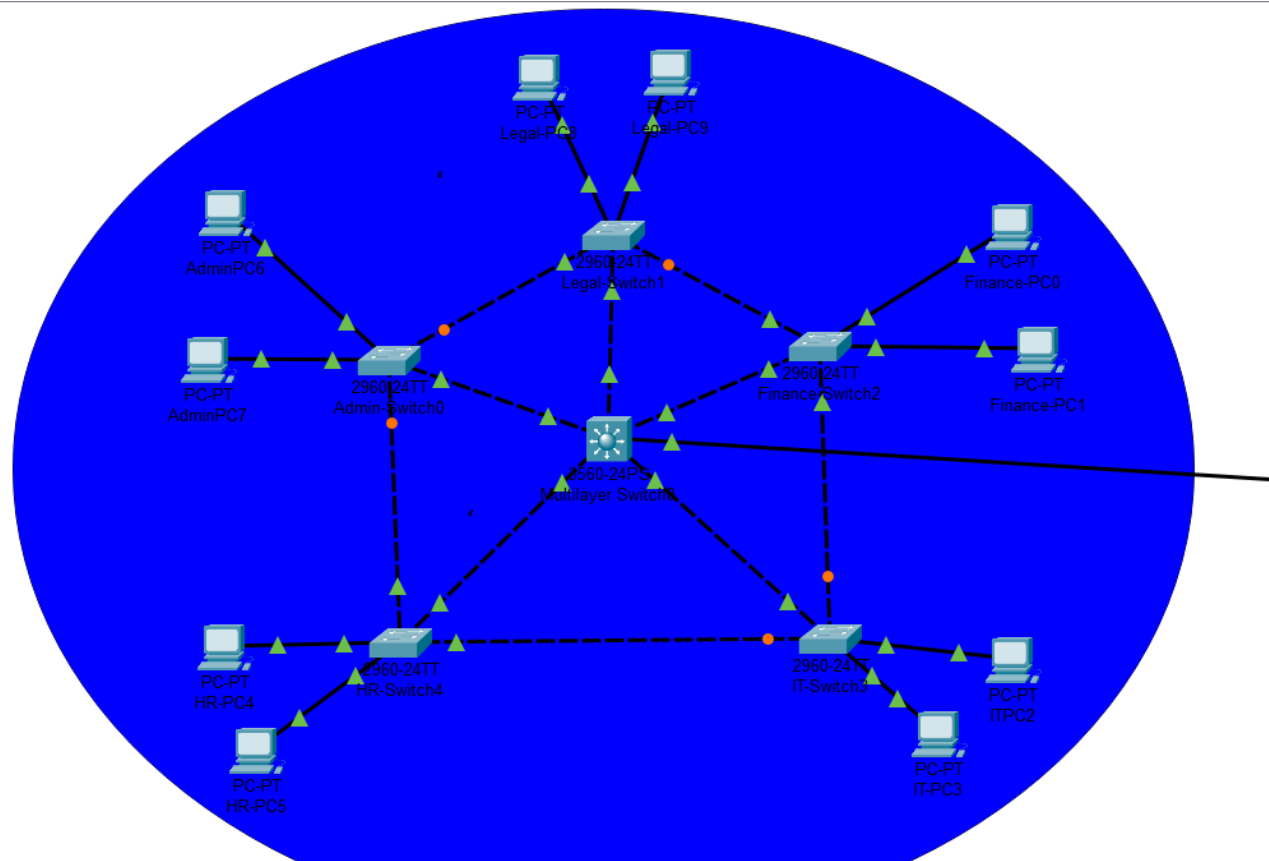
b) The ip interfaces of in the router

```
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0/0.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/0/0.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/0/0.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/0/0.40	192.168.40.1	YES	manual	up	up
GigabitEthernet0/0/0.50	192.168.50.1	YES	manual	up	up
GigabitEthernet0/0/1	unassigned	YES	manual	up	up
GigabitEthernet0/0/1.60	192.168.60.1	YES	manual	up	up
GigabitEthernet0/0/2	unassigned	YES	unset	up	down
Vlan1	unassigned	YES	unset	up	down

```
Router#
```

c) Logical design of the hybrid topology:



d) The ACL placed in the router:

```
Router#show access-list
Extended IP access list 100
 10 permit tcp any any eq www
 20 permit tcp any any eq domain
 30 permit ip 192.168.10.0 0.0.0.255 192.168.60.0 0.0.0.255
 40 permit ip 192.168.20.0 0.0.0.255 192.168.60.0 0.0.0.255
 50 permit ip 192.168.30.0 0.0.0.255 192.168.60.0 0.0.0.255
 60 permit ip 192.168.40.0 0.0.0.255 192.168.60.0 0.0.0.255
 70 permit ip 192.168.50.0 0.0.0.255 192.168.60.0 0.0.0.255
 80 deny ip any 192.168.60.0 0.0.0.255
```

11. Advantages of the setup:

- Centralized server access
- Departmental segmentation
- Easier network management

12. Disadvantages of the setup:

- Single point of failure in the star (server farm)
- Slightly complex interconnection.

13. Conclusion:

The proposed hybrid network effectively combined both the star and the ring topology, interconnected with a router to enable inter-VLAN communication. However, in the ring topology the multilayered switch does the internal routing ensuring that there is lower latency in the ring. This design further ensures centralized access to critical resources in the server farm while maintaining departmental segmentation for security and the use of ACL. The design is scalable, manageable and secure thus providing the system with a robust foundation for current operations and future network growth.